

EX. NO: 3

IMPLEMENTATION OF HILL CIPHER

AIM:

To write a C program to implement the hill cipher substitution techniques.

ALGORITHM:

STEP-1: Read the plain text and key from the user.

STEP-2: Split the plain text into groups of length three.

STEP-3: Arrange the keyword in a 3*3 matrix.

STEP-4: Multiply the two matrices to obtain the cipher text of length three.

STEP-5: Combine all these groups to get the complete cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
int main()
```

```
{
```

```
    int a[3][3] = {
```

```
        {6, 24, 1},
```

```
        {13, 16, 10},
```

```
        {20, 17, 15}
```

```
    };
```

```
    int b[3][3] = {
```

```
        {8, 5, 10},
```

```
        {21, 8, 21},
```

```
        {21, 12, 8}
```

```
    };
```

```
    int c[3], d[3];
```

```
char msg[20];

int i, j, t;

printf("Enter Plain Text (3 letters): ");

scanf("%s", msg);

if (strlen(msg) != 3)
{
    printf("Please enter exactly 3 uppercase letters.\n");
    return 0;
}

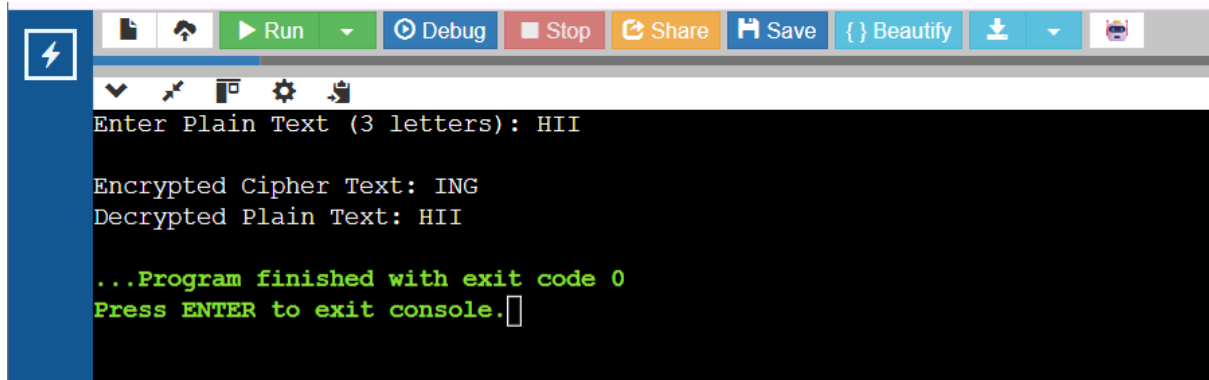
for (i = 0; i < 3; i++)
{
    if (msg[i] >= 'a' && msg[i] <= 'z')
        msg[i] = msg[i] - 32;

    c[i] = msg[i] - 'A';
}

for (i = 0; i < 3; i++)
{
    t = 0;
    for (j = 0; j < 3; j++)
    {
        t += a[i][j] * c[j];
    }
    d[i] = t % 26;
}
```

```
printf("\nEncrypted Cipher Text: ");  
for (i = 0; i < 3; i++)  
{  
    printf("%c", d[i] + 'A');  
}  
  
for (i = 0; i < 3; i++)  
{  
    t = 0;  
    for (j = 0; j < 3; j++)  
    {  
        t += b[i][j] * d[j];  
    }  
    c[i] = t % 26;  
}  
  
printf("\nDecrypted Plain Text: ");  
for (i = 0; i < 3; i++)  
{  
    printf("%c", c[i] + 'A');  
}  
  
return 0;  
}
```

OUTPUT:



The screenshot shows a code editor interface with a toolbar at the top containing buttons for Run, Debug, Stop, Share, Save, Beautify, and a download icon. Below the toolbar is a dark-themed console window with the following text:

```
Enter Plain Text (3 letters): HII
Encrypted Cipher Text: ING
Decrypted Plain Text: HII
...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT:

Thus the hill cipher substitution technique had been implemented successfully in C.